

27.3 Principal component regression

Ridge regression keeps every principal direction and shrinks each by a factor between zero and one. Principal component regression makes the factors zero or one: it keeps the directions in which the regressors vary most and discards the rest. Principal components (Hotelling 1933) are an old remedy for collinearity (Massy 1965), and Proposition 10.4.3 computed the bias and variance of regression on the leading ones in uncentred form. This section proves when it beats least squares and makes precise the warning that goes with it.

27.3.1 Definition

Keep the conventions of Section 27.2: $\mathbf{X}$ is $n \times p$, centred, of full column rank, with its columns on a common scale, and $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ with $d_1 \geq \dots \geq d_p > 0$.

Definition 27.3.1. Principal component regression

The **principal components** of the regressors are the vectors $\mathbf{z}_j = \mathbf{X}\mathbf{v}_j = d_j\mathbf{u}_j$, $j = 1, \dots, p$. Their sample variances are $d_j^2/(n-1)$, they are mutually orthogonal, and $\mathbf{v}_j$ is the j th **loading vector**. For $0 \leq k \leq p$, the **principal component regression** estimate with k components is

$$\hat{\boldsymbol{\beta}}_k^{\text{PC}} = \sum_{j \leq k} \hat{\alpha}_j \mathbf{v}_j = \mathbf{V}_k \mathbf{V}_k^\top \hat{\boldsymbol{\beta}}, \quad \mathbf{V}_k = [\mathbf{v}_1, \dots, \mathbf{v}_k],$$

obtained by regressing $\mathbf{y}$ on $\mathbf{z}_1, \dots, \mathbf{z}_k$ and transforming the coefficients back.

Because the $\mathbf{z}_j$ are orthogonal, the coefficient of $\mathbf{z}_j$ in the regression on $\mathbf{z}_1, \dots, \mathbf{z}_k$ is $\mathbf{z}_j^\top \mathbf{y} / \|\mathbf{z}_j\|^2 = c_j/d_j = \hat{\alpha}_j$, the same for every $k \geq j$; the regression on the leading components is least squares with the trailing canonical coordinates set to zero. The shrinkage factors are $f_j = 1$ for $j \leq k$ and $f_j = 0$ for $j > k$.

Proposition 27.3.1. Bias and variance of principal component regression

Let $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$, $\boldsymbol{\alpha} = \mathbf{V}^\top \boldsymbol{\beta}$ and $\tau_j^2 = d_j^2 \alpha_j^2 / \sigma^2$.

- (a) $\hat{\boldsymbol{\beta}}_k^{\text{PC}}$ is the least squares estimate subject to the $p - k$ restrictions $\mathbf{v}_j^\top \mathbf{b} = 0$, $j > k$. Its fitted vector is $\mathbf{M}_k \mathbf{y}$, where $\mathbf{M}_k = \sum_{j \leq k} \mathbf{u}_j \mathbf{u}_j^\top$ is the orthogonal projection onto $\text{span}(\mathbf{z}_1, \dots, \mathbf{z}_k)$.
- (b) $E \hat{\boldsymbol{\beta}}_k^{\text{PC}} - \boldsymbol{\beta} = -\sum_{j > k} \alpha_j \mathbf{v}_j$ and $\text{Cov}(\hat{\boldsymbol{\beta}}_k^{\text{PC}}) = \sigma^2 \sum_{j \leq k} d_j^{-2} \mathbf{v}_j \mathbf{v}_j^\top$. The total mean squared error is $\sum_{j > k} \alpha_j^2 + \sigma^2 \sum_{j \leq k} d_j^{-2}$, and the mean squared error of the fitted values is $E \|\mathbf{X} \hat{\boldsymbol{\beta}}_k^{\text{PC}} - \mathbf{X} \boldsymbol{\beta}\|^2 = k\sigma^2 + \sum_{j > k} d_j^2 \alpha_j^2$.
- (c) $\hat{\boldsymbol{\beta}}_k^{\text{PC}}$ is at least as good as $\hat{\boldsymbol{\beta}}$ in the matrix sense iff $\sum_{j > k} \tau_j^2 \leq 1$.
- (d) Among all estimators $\tilde{\boldsymbol{\beta}}_{\mathbf{f}}$ with every $f_j \in \{0, 1\}$, both mean squared errors in
 - (a) are minimized by keeping exactly the directions with $\tau_j^2 > 1$ (ties may go either way). Principal component regression with k components is this best selection iff $\tau_j^2 \geq 1$ for $j \leq k$ and $\tau_j^2 \leq 1$ for $j > k$.

Proof.

- (a) The restrictions say that $\mathbf{b} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, that is, $\mathbf{b} = \mathbf{V}_k \mathbf{g}$. Then $\mathbf{Xb} = \mathbf{U}_k \mathbf{D}_k \mathbf{g}$ with $\mathbf{U}_k, \mathbf{D}_k$ the leading blocks, and $\|\mathbf{y} - \mathbf{U}_k \mathbf{D}_k \mathbf{g}\|^2$ is minimized at $\mathbf{D}_k \mathbf{g} = \mathbf{U}_k^\top \mathbf{y}$, so $\mathbf{g} = (\hat{\alpha}_1, \dots, \hat{\alpha}_k)^\top$ and the fit is $\mathbf{U}_k \mathbf{U}_k^\top \mathbf{y} = \mathbf{M}_k \mathbf{y}$.
- (b) The moments are Proposition 10.4.3, and the mean squared errors are Proposition 27.1.2(a) and (b) with the factors 0 and 1.
- (c) By Theorem 27.1.1(c) with the saving $\Delta_{\mathbf{A}} = \sigma^2 \sum_{j>k} d_j^{-2} \mathbf{v}_j \mathbf{v}_j^\top$, which is nonnegative definite with column space $\text{span}(\mathbf{v}_{k+1}, \dots, \mathbf{v}_p)$, and bias $\delta = -\sum_{j>k} \alpha_j \mathbf{v}_j$, which lies in that column space. The Moore–Penrose inverse is $\Delta_{\mathbf{A}}^+ = \sigma^{-2} \sum_{j>k} d_j^2 \mathbf{v}_j \mathbf{v}_j^\top$, so $\delta^\top \Delta_{\mathbf{A}}^+ \delta = \sum_{j>k} d_j^2 \alpha_j^2 / \sigma^2$.
- (d) The losses are sums over directions, and by Proposition 27.1.2(d) the j th term is smaller with $f_j = 0$ than with $f_j = 1$ iff $\tau_j^2 < 1$.

■

By part (c), discarding directions improves the estimate of *every* linear combination exactly when the total signal they carry, in noise units, is at most one. Part (b) shows that the two losses charge a dropped direction differently: for β it costs α_j^2 and saves σ^2/d_j^2 , which is large when d_j is small; for the mean it costs $d_j^2 \alpha_j^2$ and saves only σ^2 . Collinearity is a problem for coefficients, not for fitted values (Proposition 26.4.1).

27.3.2 Low-variance components need not be irrelevant

Part (d) is the crux. The right directions to drop are those with small $\tau_j^2 = d_j^2 \alpha_j^2 / \sigma^2$; principal component regression drops those with small d_j^2 . The two orderings agree only if the α_j do not grow too fast as d_j shrinks, and nothing in the model says they should: the components are computed from $\mathbf{X}$ alone. Jolliffe (1982) gave published examples in which low-variance components were important predictors, and Hadi and Ling (1998) showed how a component with a tiny variance can carry most of a regression (Exercise (B2)).

The signal-to-noise ratios can be estimated: under normal errors $c_j^2/\sigma^2 - 1$ is unbiased for τ_j^2 , and the component t statistics $t_j = c_j/s$ estimate τ_j (Exercise (B1)). Part (d) then suggests keeping the components with $t_j^2 > 2$. Such a supervised selection is not a fixed projection, and the leave-one-out and standard-error formulas do not apply to it without adjustment (Chapter 29).

Example. Principal components of Longley's regressors

The six standardized Longley regressors have principal components whose shares of the total variance are 76.723%, 19.589%, 3.390%, 0.249%, 0.043% and 0.006%, so the usual rules of thumb keep two or three. The component t statistics are 42.66, 5.40, 10.66, -0.55 , 3.96 and 1.72 (Figure 27.3.1(a)). The fifth component, with less than a two-thousandth of the variance, is nearly as strongly related to employment as the second, while the fourth, with more variance than the fifth and sixth together, is irrelevant.

Because principal component regression fits a projection (Proposition 27.3.1(a)), the leave-one-out prediction errors follow from Theorem 20.3.1 with the leverages $1/n + \sum_{j \leq k} u_{ij}^2$. The root mean squared leave-one-out errors for $k = 0, 1, \dots, 6$ are 3627, 1109, 1067, 506, 606, 435 and 425, in thousands. Adding the fourth component makes predictions *worse*, and adding the fifth makes them much better; the best number of leading components is 6, which

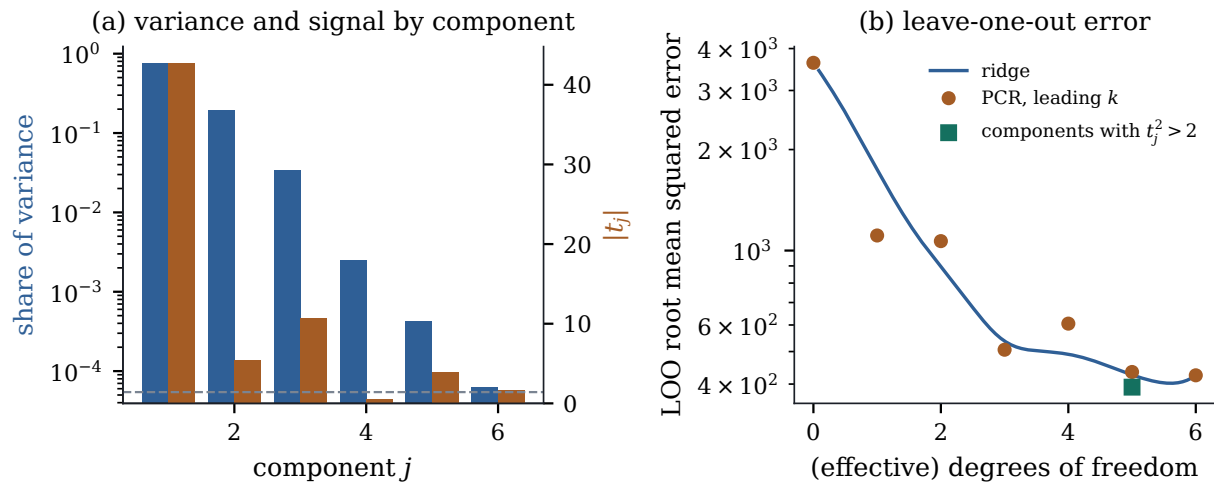


Figure 27.3.1: Principal components of Longley's regressors. (a) Share of the regressors' variance (log scale, left) and the absolute component t statistic (right; dashed line at $\sqrt{2}$). (b) Leave-one-out root mean squared prediction error of principal component regression with the leading k components, of ridge regression against its effective degrees of freedom, and of the components with $t_j^2 > 2$, each for a fixed choice of components or λ .

is least squares itself. The rule $t_j^2 > 2$ keeps components 1, 2, 3, 5, 6. Applied once to all sixteen years and then treated as fixed, it gives 392, but that figure ignores the selection. Redoing the selection within each fold (*nested* leave-one-out) gives 444, worse than least squares. Searching all 64 subsets of components picks 1, 2, 3, 5 with 389, optimistic for the same reason. With three leading components the coefficients are 1023, 1260, -1085 , -417 , 1070 and 966: all six regressors are retained, with the GNP and YEAR coefficients now positive and comparable.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
names = ["GNPDEFL", "GNP", "UNEMP", "ARMED", "POP", "YEAR"]
Z = data[names].to_numpy()
y = data["TOTEMP"].to_numpy()
n, p = Z.shape
X = (Z - Z.mean(axis=0)) / Z.std(axis=0, ddof=1)
yc = y - y.mean()
U, d, Vt = np.linalg.svd(X, full_matrices=False)
c = U.T @ yc # c_j = u_j^T y, the fitted
              coordinate of component j

def fit_components(keep):
    """Least squares on the chosen component scores: coefficients and leave-one-out
    error."""
    keep = list(keep)
    beta = Vt[keep].T @ (c[keep] / d[keep])
    resid = yc - X @ beta
    h = 1 / n + np.sum(U[:, keep] ** 2, axis=1) # leverages, intercept
    included
    return beta, np.mean((resid / (1 - h)) ** 2)

resid_ls = yc - U @ c
s = np.sqrt(resid_ls @ resid_ls / (n - p - 1))
t = c / s # t statistic of each component
```

```

share = d ** 2 / np.sum(d ** 2)                                # share of the regressors' variance
for k in range(p + 1):
    beta_k, loo_k = fit_components(range(k))
    print(f"k = {k}  L00 = {loo_k:9.0f}  coefficients", np.round(beta_k, 0))
print("variance shares:", np.round(share, 5))
print("component t statistics:", np.round(t, 2))

```

[Run this code online](#)

27.3.3 Ridge and principal components compared

Both methods order the directions by d_j alone, ridge softly and principal components hard, and neither dominates. If the trailing canonical coefficients are zero, principal components with the right k is close to the oracle, while ridge must leave variance in the trailing directions or bias the leading ones; if all the α_j^2 are equal, ridge with $\lambda = \sigma^2/\alpha_j^2$ is the oracle and no zero-one choice matches it ([Exercise \(B3\)](#)). In the simulations of Frank and Friedman (1993) ridge was generally at least as good as principal components. For Longley's data neither method gains over least squares (leading components never do, and tuned ridge has nested leave-one-out error 441 against 425), because the fifth direction matters and both treat it as close to noise. Partial least squares, which builds its directions from $\mathbf{y}$ as well as $\mathbf{X}$, is one response ([Exercise \(C1\)](#)).

27.3.4 Exercises

27.3.4.1 A. Check your understanding

Exercise. A1

Show that for $k < p$ the estimate $\hat{\boldsymbol{\beta}}_k^{\text{PC}}$ generally has no zero coordinates, so that principal component regression reduces dimension without selecting variables. When does it set the coefficient of regressor j to zero?

Exercise. A2

Show that the fitted values of principal component regression with k components, with an unpenalized intercept, are $(n^{-1}\mathbf{1}\mathbf{1}^\top + \mathbf{M}_k)\mathbf{y}$, that this matrix is an orthogonal projection of rank $k + 1$, and that its leverages are $1/n + \sum_{j \leq k} u_{ij}^2$. Why does Theorem 20.3.1 apply to principal component regression but not to ridge regression?

27.3.4.2 B. Practice

Exercise. B1

Under the normal linear model with $\text{rank}(\mathbf{X}) = p < n - 1$ (centred data), show that the $c_j = \mathbf{u}_j^\top \mathbf{Y}$ are independent $N(d_j \alpha_j, \sigma^2)$, independent of the residual sum of squares, and that $t_j = c_j/s$ has a noncentral t distribution with $n - p - 1$ degrees of freedom and noncentrality τ_j . Show that $c_j^2/\sigma^2 - 1$ is unbiased for τ_j^2 .

Solution. $\mathbf{U}^\top \mathbf{Y}$ is normal with mean $\mathbf{U}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{D} \mathbf{V}^\top \boldsymbol{\beta} = \mathbf{D} \boldsymbol{\alpha}$ and covariance $\sigma^2 \mathbf{U}^\top \mathbf{U} = \sigma^2 \mathbf{I}$, so the c_j are independent with the stated laws. They are functions of the fitted values

$\mathbf{MY} = \mathbf{UU}^\top \mathbf{Y}$, which are independent of the residual sum of squares (Theorem 7.5.1(c)); with the intercept removed by centring the residual degrees of freedom are $n - p - 1$. Hence $t_j = (c_j/\sigma)/(s/\sigma)$ is a normal with mean τ_j (up to sign) divided by an independent $\sqrt{\chi^2(n-p-1)/(n-p-1)}$, which is the noncentral t (Definition 4.1.3). Finally $E(c_j^2/\sigma^2) = \text{Var}(c_j/\sigma) + \{E(c_j/\sigma)\}^2 = 1 + \tau_j^2$.

Exercise. B2

Let x_1, x_2 be standardized regressors with correlation $\rho = 0.99$ and let $E(Y) = x_1 - x_2$. Find the canonical coefficients and show that $\alpha_1 = 0$. Compute τ_1^2 and τ_2^2 with $n = 100$, unit-variance columns and $\sigma = 1$. Which principal component regression is best, and what does a one-component principal component regression estimate?

Solution. With $\mathbf{X}^\top \mathbf{X} = (n-1)\mathbf{R}$, the eigenvectors are $\mathbf{v}_1 = (1, 1)^\top/\sqrt{2}$ and $\mathbf{v}_2 = (1, -1)^\top/\sqrt{2}$, with $d_1^2 = (n-1)(1+\rho)$ and $d_2^2 = (n-1)(1-\rho)$. For $\boldsymbol{\beta} = (1, -1)^\top$, $\alpha_1 = \mathbf{v}_1^\top \boldsymbol{\beta} = 0$ and $\alpha_2 = \sqrt{2}$. So $\tau_1^2 = 0$ and $\tau_2^2 = (n-1)(1-\rho) \cdot 2/\sigma^2 = 99 \times 0.01 \times 2 = 1.98$. By [Proposition 27.3.1\(d\)](#), the best zero-one selection keeps only the *second* component. Principal component regression with one component keeps the first, estimates $\alpha_1 = 0$ and has expected value $\mathbf{0}$: it discards the whole regression.

Exercise. B3

Consider a design with two principal directions. (i) If $\alpha_2 = 0 \neq \alpha_1$, show that the oracle factors of Proposition 27.1.2(c) are $(\tau_1^2/(1+\tau_1^2), 0)$, that no ridge estimate attains them, and that principal component regression with $k = 1$ exceeds the oracle's total mean squared error by only $(\sigma^2/d_1^2)/(1+\tau_1^2)$. (ii) If $\alpha_1^2 = \alpha_2^2 > 0$, show that ridge with $\lambda = \sigma^2/\alpha_1^2$ attains the oracle, and that every principal component regression is strictly worse.

Solution. Part (i): $\tau_2^2 = 0$ gives $f_2^* = 0$. Ridge has $f_2(\lambda) = d_2^2/(d_2^2 + \lambda) > 0$ for every finite λ , so it never attains $f_2^* = 0$. Principal components with $k = 1$ has factors $(1, 0)$. By Proposition 27.1.2(a) its first term is σ^2/d_1^2 against the oracle's $(\sigma^2/d_1^2)\tau_1^2/(1+\tau_1^2)$, and the second terms are both zero; the difference is $(\sigma^2/d_1^2)/(1+\tau_1^2)$. (ii) With $\alpha_1^2 = \alpha_2^2 = a$ and $\lambda = \sigma^2/a$, ridge has factors $d_j^2/(d_j^2 + \sigma^2/a) = d_j^2 a/(d_j^2 a + \sigma^2) = \tau_j^2/(1+\tau_j^2)$, the oracle. Each term of Proposition 27.1.2(a) is a strictly convex quadratic in f_j , so the oracle is its unique minimizer, and since $0 < \tau_j^2/(1+\tau_j^2) < 1$, every choice of factors in $\{0, 1\}$ is strictly worse.

27.3.4.3 C. Going deeper

Exercise. C1

The first direction of **partial least squares** is $\mathbf{w}_1 \propto \mathbf{X}^\top \mathbf{y}$, the direction maximizing the sample covariance of $\mathbf{X}\mathbf{w}$ with $\mathbf{y}$ over unit vectors $\mathbf{w}$. Show that $\mathbf{w}_1 \propto \sum_j d_j c_j \mathbf{v}_j$, and compare with the first principal direction $\mathbf{v}_1$. Show that the one-dimensional fit along $\mathbf{w}_1$ never ignores a direction j with $c_j \neq 0$.